

# Zaros Finance Audit Summary

Prepared by Cyfrin

Version 2.0


**Lead Auditors**

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# 1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

# 2 About This Document

This document is a public summary of the security audit Cyfrin performed for Zaros on Zaros Finance. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated July 13, 2024, was delivered to Zaros, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

# 3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

# 4 Copyright and License

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# 5 Audit Overview

Over the course of 16 days, the Cyfrin team conducted an audit on the Zaros Finance code provided by Zaros. The findings consist of 5 Critical, 3 High, 12 Medium & 12 Low severity issues with the remainder being informational and gas optimizations.

**Summary**

| | |
|---|---|
| Project Name | Zaros Finance |
| Repository | zaros-core-audit |
| Commit | de09d030c780... |
| Audit Timeline | May 30th - June 20th 2024 |
| Methods | Manual Review |

**Issues Found**

| Severity | Found | Resolved | Acknowledged |
|---|---|---|---|
| Critical Risk | 5 | 5 | 0 |
| High Risk | 3 | 3 | 0 |
| Medium Risk | 12 | 10 | 2 |
| Low Risk | 12 | 10 | 2 |
| Informational | 7 | 6 | 1 |
| Gas Optimizations | 24 | 23 | 1 |
| Total | 63 | 57 | 6 |

# 6 Audit Scope

The following contracts were included in the scope for this audit:

```
src/account-nft/AccountNFT.sol
src/account-nft/interfaces/IAccountNFT.sol
src/external/chainlink/ChainlinkUtil.sol
src/external/chainlink/interfaces/IAggregatorV3.sol
src/external/chainlink/interfaces/IAutomationCompatible.sol
src/external/chainlink/interfaces/IFeeManager.sol
src/external/chainlink/interfaces/ILogAutomation.sol
src/external/chainlink/interfaces/IStreamsLookupCompatible.sol
src/external/chainlink/interfaces/IVerifierProxy.sol
src/external/chainlink/keepers/BaseKeeper.sol
src/external/chainlink/keepers/liquidation/LiquidationKeeper.sol
src/external/chainlink/keepers/market-order/MarketOrderKeeper.sol
src/perpetuals/PerpsEngine.sol
src/perpetuals/branches/GlobalConfigurationBranch.sol
src/perpetuals/branches/LiquidationBranch.sol
src/perpetuals/branches/OrderBranch.sol
src/perpetuals/branches/PerpMarketBranch.sol
src/perpetuals/branches/SettlementBranch.sol
src/perpetuals/branches/TradingAccountBranch.sol
src/perpetuals/leaves/FeeRecipients.sol
src/perpetuals/leaves/GlobalConfiguration.sol
src/perpetuals/leaves/MarginCollateralConfiguration.sol
src/perpetuals/leaves/MarketOrder.sol
src/perpetuals/leaves/OrderFees.sol
src/perpetuals/leaves/PerpMarket.sol
src/perpetuals/leaves/Position.sol
src/perpetuals/leaves/SettlementConfiguration.sol
src/perpetuals/leaves/TradingAccount.sol
src/tree-proxy/RootProxy.sol
src/tree-proxy/branches/LookupBranch.sol
src/tree-proxy/branches/UpgradeBranch.sol
src/tree-proxy/leaves/Branch.sol
src/tree-proxy/leaves/LookupTable.sol
src/tree-proxy/leaves/RootUpgrade.sol
src/usd/USDToken.sol
src/utils/Constants.sol
src/utils/Errors.sol
src/utils/Math.sol
```

# 7 Risk Classification

| | **Impact: High** | **Impact: Medium** | **Impact: Low** |
|---|---|---|---|
| **Likelihood: High** | Critical | High | Medium |
| **Likelihood: Medium** | High | Medium | Low |
| **Likelihood: Low** | Medium | Low | Low |

# 8 Protocol Summary

Zaros Finance is an on-chain de-centralized Perpetuals exchange aiming to connect Liquid Re(Staking) Tokens (LSTs & LRTs) with Perpetual Futures, letting liquidity providers use their LSTs and LRTs to provide liquidity earning additional yield.

Zaros features:

- a native overcollateralized stablecoin USDz backed by the deposited LSTs/LRTs used by the protocol to settle trading positions
- the Tree Proxy Pattern a novel approach to modular smart contract proxies
- cross-margin trading accounts which allow users to segregate trading strategies into separate trading accounts, each backed by its own set of collateral
- trading on non-crypto markets including select foreign exchange pairs and commodities

# 9 Attestation

Cyfrin attests that it performed a security review of the Zaros Finance code provided by Zaros.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at `https://github.com/Cyfrin/cyfrin-audit-reports`. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2024-07-13-cyfrin-zaros-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Zaros has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=1a07e1ae79bf108c6e100f412e8aaec08003c4878ac5bd55b2906c4dc47a44d7
F=2024-07-13-cyfrin-zaros-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.